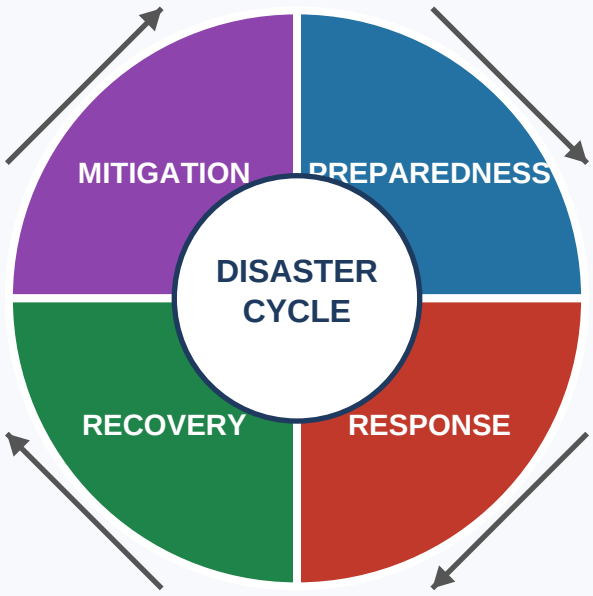
DISASTER MANAGEMENT

Cycle | Phases | Global Frameworks | India's System | Science & Technology

1. THE DISASTER MANAGEMENT CYCLE

Mitigation - Reduce the risk before a disaster

Hazard zoning, safe building codes, land-use planning, retrofitting (structural and non-structural measures).



Preparedness - Be ready to respond

Early warning systems, evacuation plans, mock drills, training, awareness, stock of relief material.

Recovery - Rebuild after the disaster

Damage and needs assessment, restoring critical infrastructure, rehabilitation and 'build back better'.

Response - Act during the disaster

Evacuation, search and rescue, emergency operation centre, medical aid, relief and communication.

2. ACTIVITIES IN EACH PHASE

PRE-DISASTER

* Risk assessment and analysis
* Risk mapping, zonation and micro-zonation
* Prevention and mitigation
* Early warning systems
* Preparedness and capacity development
* Awareness and training

DURING DISASTER

* Evacuation
* Disaster communication
* Search and rescue
* Emergency Operation Centre (EOC)
* Incident Command System (ICS)
* Relief: food, water, shelter, medical aid

POST-DISASTER

* Damage and needs assessment
* Restoration of critical infrastructure
* Early recovery and rehabilitation
* Reconstruction and redevelopment
* Lessons learnt and review of plans
* Psychosocial support

3. GLOBAL FRAMEWORKS FOR DISASTER RISK REDUCTION

1990-1999 — IDNDR

UN International Decade for Natural Disaster Reduction

1994 — Yokohama Strategy

Strategy and Plan of Action for a Safer World; stressed prevention and preparedness

2005-2015 — Hyogo Framework

Framework for Action: building the resilience of nations and communities

2015-2030 — Sendai Framework

Understand risk; strengthen governance; invest in resilience; improve preparedness and 'build back better'

4. DISASTER MANAGEMENT IN INDIA

* Disaster Management Act, 2005 - legal basis; National Policy on Disaster Management, 2009.
* NDMA (apex body, chaired by the Prime Minister) lays down policies, plans and guidelines.
* SDMA (chaired by the Chief Minister) and DDMA (chaired by the District Collector/Magistrate) plan and act at state and district level.
* NDRF - specialised response force; IMD and CWC - weather, cyclone and flood warnings.
* NGOs, community volunteers and inter-governmental agencies support relief and recovery.
* Gujarat: GSDMA, formed after the 2001 earthquake.

5. SCIENCE & TECHNOLOGY (RS, GIS, GPS)

* Remote Sensing (RS): satellite images to monitor floods, cyclones and damage.
* GIS: hazard, vulnerability and risk maps; planning of relief routes and shelters.
* GPS: locating affected areas and guiding rescue teams.
* Early warning and its dissemination: sirens, SMS, radio and TV alerts.
* Land-use planning and development regulations; disaster-safe design and construction.
* S&T institutions: IMD, ISRO, NDMA, NIDM and state institutes.

6. MITIGATION MEASURES

STRUCTURAL

Earthquake-resistant buildings, cyclone shelters, embankments and dams, flood-proof roads, retrofitting of weak structures.

NON-STRUCTURAL

Laws and building codes, hazard zoning and land-use planning, awareness and training, insurance, early warning and community drills.

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Sources: Disaster Management Act 2005; NDMA; UNDRR (Sendai Framework); Course syllabus METXX11501.